

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT2 — DATA ANALYSIS PROJECT

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Analysis (BL4)	Assessment:	Assessment Tool 2 – Data Analysis Project
Weightage:	40% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Analyze real datasets using ECDFs, skewed-distribution techniques, Q–Q plots, multi-distribution comparisons, and proportion visualizations to identify patterns and justify conclusions.(BL4)</i>		
Student Name:	RAGUL.M	Reg. No.:	192524066
Section / Batch:	SLOT C-DSA0606	Date:	18/08/2026

Code of Conduct

I RAGUL.M/192524066 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: RAGUL.M

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond construction: analyze the given data, identify the relevant pattern, and justify your conclusion.
- Show all supporting calculations (e.g., IQR, percentiles, angle calculations) where relevant.
- Where a question asks you to compare or recommend, state your reasoning explicitly — a correct chart choice without justification will not receive full marks.

Annexure A – DATA ANALYSIS PROJECT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Data Analysis Project

CO3 · BT4: Analyze ·

This rubric is used to assess student responses to the CO3-AT2 Data Analysis Project, in which students analyze a given dataset using ECDF, skewed-distribution, Q–Q plot, multi-distribution, or proportion-visualization techniques and justify their conclusions.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant technique (ECDF, skewed-distribution/log transform, Q–Q plot, multi-distribution comparison, or proportion visualization) and correctly identifies which pattern applies to the given data.	Good understanding of the relevant technique, with only minor gaps in identifying the correct pattern.	Basic understanding, but with some misconceptions about which technique or pattern applies to the data.	Poor understanding of core concepts; misidentifies the pattern or technique required by the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, going beyond description to explain why the pattern occurs or what it implies.	Provides clear analysis with adequate reasoning tied to the data, for most parts.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies analytical techniques to compare, justify, or recommend a conclusion that is well-supported by the data and any calculations shown.	Applies analytical techniques correctly, with only minor errors in reasoning or calculation.	Limited application; conclusions are only partially supported by the data or calculations.	Fails to apply appropriate analytical techniques; conclusions are unsupported or incorrect.
Organization	15%	The response is	The response is	Basic structure;	Poorly organized;

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		clearly structured, addressing every sub-part (a–b) in a logical sequence with coherent overall flow.	organized and addresses each sub-part, with only minor issues in flow or clarity.	sub-parts are addressed but the response lacks clarity or a logical sequence.	the response is incoherent, and sub-parts are left unaddressed or answered out of order.
Accuracy	10%	All parts of the answer — pattern identification, calculations, and conclusions — are completely accurate and well-suited to the data.	Mostly accurate, with only minor omissions or a small error in one sub-part.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple sub-parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

P4 — ECDF ANALYSIS

Introduction

The Empirical Cumulative Distribution Function (ECDF) is used to study the distribution of observations across different values. It represents the cumulative proportion of observations that are less than or equal to a particular value. Comparing two ECDF curves helps in understanding how two groups differ throughout their entire distributions.

Objective

The objective is to analyze the performance distributions of Class X and Class Y and understand the meaning of the crossing pattern between their ECDF curves.

Analysis

The ECDF curves of Class X and Class Y intersect at different score levels. This crossing pattern indicates that neither class consistently performs better across the complete range of scores. At some score ranges, one class may have a better distribution, while at other ranges the second class may perform better.

If one ECDF curve remained consistently on one side of the other, it would indicate a stronger form of distributional dominance. Since the curves cross, the relative performance changes depending on the score range.

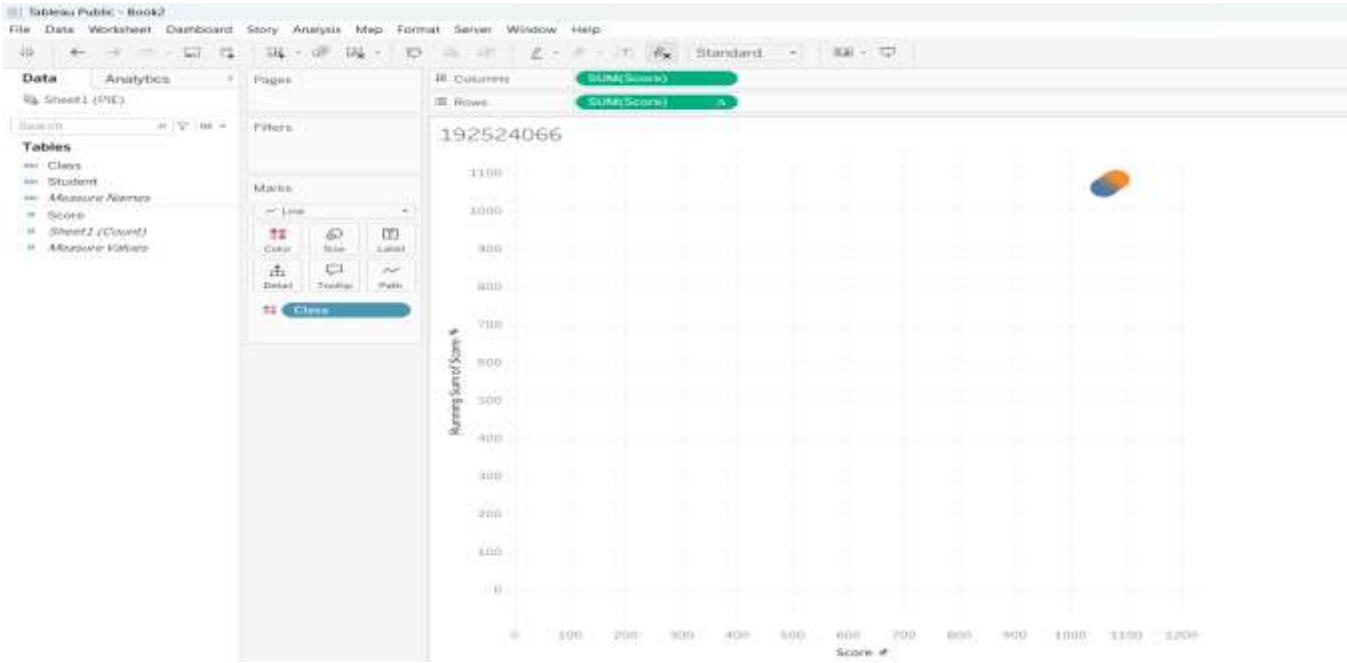
The class requiring additional support should be identified by examining the score ranges where its ECDF indicates a higher concentration of lower scores. Therefore, support should not be decided using only the overall average performance.

Result

The ECDF comparison shows that Class X and Class Y have different performance patterns across the score range, with neither class completely dominating the other.

Conclusion

The crossing ECDF pattern demonstrates that distributional comparison provides more information than simply comparing average scores. ECDF is therefore useful for identifying specific score ranges where students may require additional academic support.



P10 — SKEWED DISTRIBUTION AND LOG TRANSFORMATION

Introduction

Skewed distributions occur when data is not evenly distributed around its center. A right-skewed distribution contains a long tail toward larger values. Log transformation is commonly used for datasets with very large differences between observations because it compresses extreme values and makes the distribution easier to interpret.

Objective

The objective is to analyze the distribution of online video views, identify its skewness and power-law-like behavior, and explain the usefulness of logarithmic transformation.

Analysis

The video-view distribution is strongly right-skewed. Most videos receive relatively small or moderate numbers of views, while a few videos receive extremely large numbers of views. This creates a long right tail.

The rapid increase in views suggests a heavy-tailed or power-law-like pattern, where a small number of observations account for a very large portion of the total activity.

A linear-scale visualization can be misleading because the extremely popular videos dominate the scale and make smaller values appear almost insignificant. A logarithmic transformation compresses the large values while expanding the visual space available for smaller values.

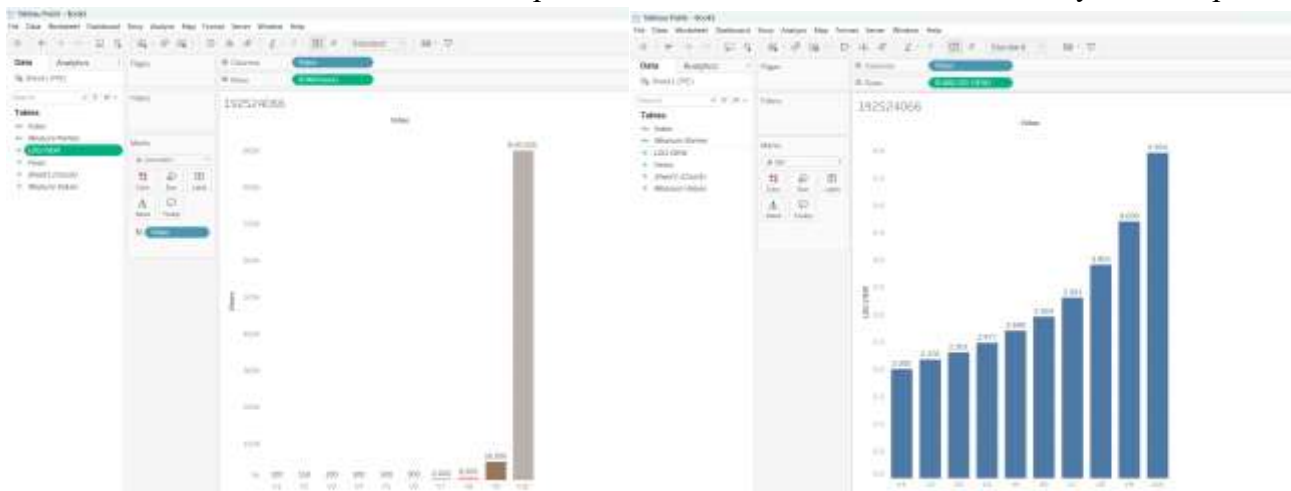
For example, the difference between 100 views and 900,000 views is enormous on the original scale. After applying a logarithmic scale, the difference becomes much easier to display and compare.

Result

The analysis identifies the dataset as highly right-skewed with a heavy-tailed pattern. Log transformation provides a clearer representation of both small and large observations.

Conclusion

Log transformation is an effective method for analyzing highly skewed data. It reduces the visual dominance of extreme values and makes patterns within the dataset easier to identify and interpret.



P11 — Q-Q PLOT ANALYSIS

Introduction

A Q-Q plot is a statistical visualization used to compare the distribution of observed data with a theoretical distribution. When data follows a Normal distribution, the points in a Q-Q plot generally follow a straight reference line. Deviations from the line indicate differences between the observed and theoretical distributions.

Objective

The objective is to analyze the distribution of student quiz scores and determine what the deviation at the upper end of the Q-Q plot indicates.

Analysis

The Q-Q plot follows the reference line closely through the middle portion, indicating that the central part of the quiz-score distribution is reasonably consistent with a Normal distribution.

However, the curve moves sharply upward at the top-right end. This indicates that the upper tail contains unusually high observations compared with what would be expected from a Normal distribution.

This deviation may occur because a small number of students achieved exceptionally high scores. Another plausible explanation could be a scoring effect, such as a large group of students scoring within the normal range while a few high-performing students reach scores close to the maximum possible mark.

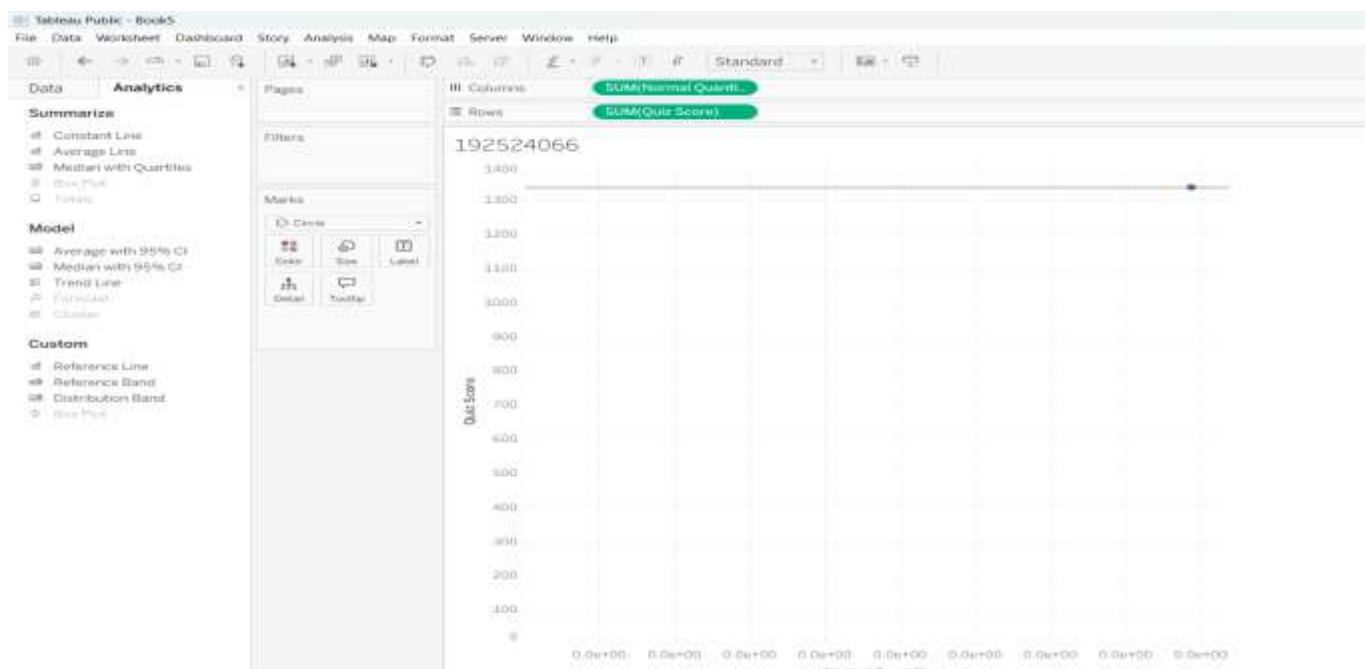
The deviation is therefore mainly associated with the upper tail rather than the entire distribution.

Result

The Q-Q plot indicates approximate Normal behavior in the central region but significant deviation at the upper end.

Conclusion

The Q-Q plot shows that the quiz-score distribution is not perfectly Normal because of unusually high values in the upper tail. This demonstrates how Q-Q plots can identify distributional differences that may not be obvious from basic numerical summaries.



P17 — COMPARISON OF MULTIPLE DISTRIBUTIONS

Introduction

Statistical comparison of distributions requires examining both central tendency and variability. Two datasets can have the same mean but completely different distributions. Therefore, visualization is important for identifying differences in spread, concentration, and consistency.

Objective

The objective is to compare two datasets with the same mean and demonstrate why the mean alone is insufficient for describing their distributions.

Analysis

Both Dataset P and Dataset Q have the same mean value of 60. However, the observations in Dataset P are concentrated closely around 60, while Dataset Q contains values spread widely from very low to very high levels.

Dataset P is therefore more consistent because its observations have very little variation. Dataset Q is much more variable because its values are widely distributed.

The range also demonstrates this difference. Dataset P has a small range, whereas Dataset Q has a substantially larger range. A box plot would clearly display this difference through the size of the interquartile range and the overall spread.

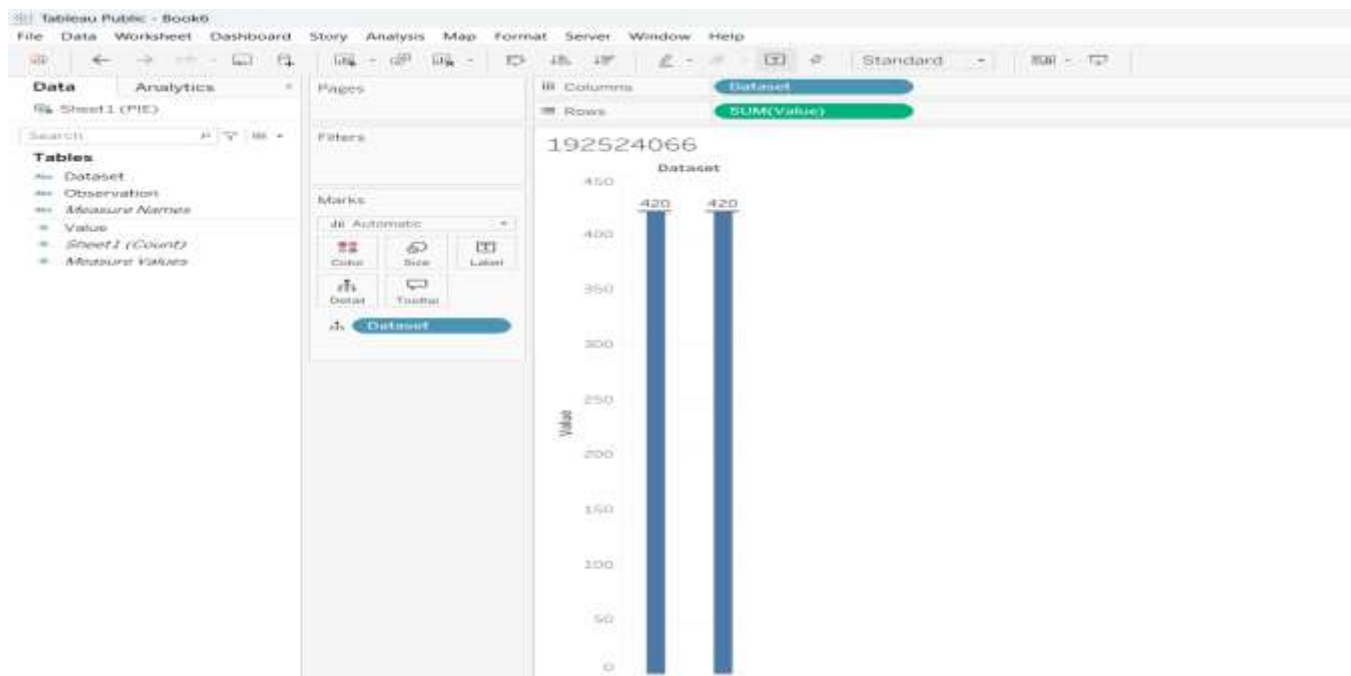
This analysis shows why comparing only the mean can lead to an incomplete or misleading interpretation.

Result

Although both datasets have the same mean, Dataset P is more consistent and Dataset Q is more variable.

Conclusion

The analysis proves that the mean alone cannot describe the complete behavior of a dataset. Measures of spread and suitable visualizations such as box plots are necessary for making a meaningful comparison between distributions.



P23 — PROPORTIONS AND FLOW VISUALIZATION

Introduction

Flow visualization is used to represent relationships between different groups or categories. In student enrollment data, it can show how students move from different schools into different colleges. Visualizing these connections makes it easier to identify major and minor flows.

Objective

The objective is to analyze the student enrollment flow between schools and colleges and identify which school contributes the largest number of students to College 1.

Analysis

College 1 receives students from all three schools. School X contributes 40 students, School Y contributes 30 students, and School Z contributes 15 students.

Therefore, School X is the largest contributor to College 1. The flow from School X to College 1 is greater than the corresponding flows from School Y and School Z.

A simple table can display the numbers, but it requires the viewer to compare several values manually. A Sankey diagram or parallel sets chart provides a clearer representation because the connections between schools and colleges can be shown visually. The thickness of each flow can represent the number of students.

This makes it easier to identify the strongest relationship and understand the overall enrollment pattern.

Result

School X contributes the highest number of students to College 1, with 40 students, followed by School Y and School Z.

Conclusion

Flow-based visualization provides a clearer understanding of relationships between schools and colleges than a simple table. A Sankey diagram or parallel sets chart can effectively communicate the direction and magnitude of student enrollment flows.

